Optimization Methods

Yuchao Feng

fengyuchao@sjtu.edu.cn

Contents

1 Optimization Problems . . . 1
1.1 Minimizers . . . 1
1.2 Optimality Conditions . . . 2
2 Line Search . . . 3
2.1 Wolfe Conditions . . . 3
2.2 Convergence of Line Search . . . 5
3 Gradient Descent . . . 6
3.1 Convex Functions . . . 6
3.2 Sublinear Rate . . . 7
3.3 Strong Convexity . . . 8
3.4 Linear Rate . . . 9
4 Newton's Method . . . 10
4.1 Pure Newton Method . . . 10
4.2 Damped Newton Method . . . 12
4.3 Hessian Modification . . . 13
4.4 Quasi-Newton Methods . . . 15

1 Optimization Problems

1.1 Minimizers

Definition 1.1. *(Optimization Problem)*
Let f_0 be a real-valued objective and let x be the optimization variable. A problem with inequality and equality constraints is

$$\min_x f_0(x) \quad \text{s.t.} \quad \begin{cases} f_i(x) \le 0 \\ h_j(x) = 0. \end{cases}$$

The functions f_i and h_j are the constraints, and $D = \{x \mid f_i(x) \le 0, h_j(x) = 0\}$ is the feasible set.

Definition 1.2. *(Global and Local Minimizers)*
A point $x^* \in D$ is a global minimizer of $f : D \to \mathbb{R}$ if $\forall x \in D, f(x^*) \leq f(x)$. It is a local minimizer if some neighborhood N of x^* satisfies $\forall x \in N \cap D, f(x^*) \leq f(x)$. The minimizer is strict if the inequality is strict for every $x \neq x^*$.

Definition 1.3. *(Semicontinuity)*
A function f is lower semicontinuous at x^* if

$$f(x^*) \leq \liminf_{x \to x^*} f(x),$$

and upper semicontinuous at x^* if $f(x^*) \geq \limsup_{x \to x^*} f(x)$.

Definition 1.4. *(Minimizing Sequence)*
Let $f^* = \inf f(x)$. A sequence $\{x_k\}$ is minimizing if $f(x_k) \to f^*$.

Proposition 1.1.
Suppose f is lower semicontinuous on D. If a minimizing sequence $\{x_k\}$ has a cluster point $x^* \in D$, then $f(x^*) = f^*$. In particular, x^* is a global minimizer.

Proof.
Along a subsequence converging to x^* one still has $f\left(x_{k_j}\right) \to f^*$. Lower semicontinuity gives $f(x^*) \leq \liminf_j f\left(x_{k_j}\right) = f^*$. The reverse inequality holds because f^* is the infimum and $x^* \in D$. ■

Remark.
A minimizing sequence need not converge to a minimizer.

1.2 Optimality Conditions

Let $D \subset \mathbb{R}^d$ be open and let $f : D \to \mathbb{R}$.

Definition 1.5. *(Gradient)*
The function f is differentiable at $x \in D$ if there is a linear form ℓ such that

$$f(x + h) = f(x) + \ell(h) + o(\|h\|) \quad (h \to 0).$$

Then $\ell(h) = \langle \nabla f(x), h \rangle$, where

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \cdots, \frac{\partial f}{\partial x_d} \right)^T$$

is the gradient of f at x.

Definition 1.6. *(Hessian)*
If the second partial derivatives exist, the Hessian $H(x) = \nabla^2 f(x)$ is the matrix

$$\left(\nabla^2 f(x)\right)_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}.$$

If $f \in C^2$, then $\partial_{ij} f = \partial_{ji} f$, namely $\nabla^2 f$ is symmetric.

Theorem 1.1. *(Taylor Expansion)*
Suppose $f \in C^2$. Then there exists $t \in (0,1)$ respectively such that

$$f(x+p) = f(x) + p^T \nabla f(x+tp),$$

$$f(x+p) = f(x) + \nabla f(x)^T p + \frac{1}{2} p^T \nabla^2 f(x+tp) p,$$

$$\nabla f(x+p) = \nabla f(x) + \int_0^1 \nabla^2 f(x+tp) p \, \mathrm{d}t.$$

Theorem 1.2. *(Necessary Conditions)*
Suppose x^* is a local minimizer of f and $\nabla^2 f$ exists and is continuous on an open neighborhood of x^*. Then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive semidefinite.

Theorem 1.3. *(Sufficient Conditions)*
Suppose $f \in C^2$ on a neighborhood of x^*. If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite, then x^* is a strict local minimizer of f.

2 Line Search

2.1 Wolfe Conditions

Definition 2.1. *(Descent Direction)*
A vector p is a descent direction for f at x if $p^T \nabla f(x) < 0$. For every sufficiently small $\varepsilon > 0$ one then has $f(x + \varepsilon p) < f(x)$.

Proposition 2.1.
Let $f(x) = \frac{1}{2}x^T Ax - b^T x$ with A symmetric and positive semidefinite, and let $p^T Ap > 0$. The exact line-search step from x along p is

$$\alpha = -\frac{\nabla f(x)^T p}{p^T Ap}, \quad \nabla f(x) = Ax - b.$$

At a point x_k, fix a descent direction p_k and write $f_k = f(x_k)$, $\nabla f_k = \nabla f(x_k)$, and $\varphi(\alpha) = f(x_k + \alpha p_k)$ for $\alpha > 0$.

Definition 2.2. *(Wolfe Conditions)*
Fix constants $0 < c_1 < c_2 < 1$. A step $\alpha > 0$ satisfies the *Armijo* condition, or sufficient-decrease condition, if

$$f(x_k + \alpha p_k) \leq f_k + c_1 \alpha \nabla f_k^T p_k.$$

It satisfies the curvature condition if

$$\nabla f(x_k + \alpha p_k)^T p_k \geq c_2 \nabla f_k^T p_k,$$

namely $\varphi'(\alpha) \geq c_2 \varphi'(0)$. The two inequalities together are the *Wolfe* conditions.

Remark.
Since $\varphi'(0) < 0$, the Armijo inequality holds for every sufficiently small $\alpha > 0$. If $\varphi'(\alpha)$ is still largely negative, a longer step can decrease f further. The curvature condition excludes those short steps.

Proposition 2.2. *(Existence)*
Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and p is a descent direction at x. If f is bounded from below on the ray $\{x + \alpha p \mid \alpha > 0\}$ and $0 < c_1 < c_2 < 1$, then the steps α that satisfy the *Wolfe* conditions form a nonempty union of open intervals.

Definition 2.3. *(Strong Wolfe Conditions)*
The strong *Wolfe* conditions keep the Armijo inequality and replace the curvature inequality by a two-sided bound: for some $c_2 \in (c_1, 1)$,

$$\left|\nabla f(x_k + \alpha p_k)^T p_k\right| \leq -c_2 \nabla f_k^T p_k.$$

Definition 2.4. *(Goldstein Condition)*
For a constant $c \in (0, 1/2)$, a step $\alpha > 0$ satisfies the *Goldstein* condition if

$$f_k + (1-c)\alpha \nabla f_k^T p_k \leq f(x_k + \alpha p_k) \leq f_k + c\alpha \nabla f_k^T p_k.$$

2.2 Convergence of Line Search

Let θ_k be the angle between p_k and $-\nabla f_k$, so

$$\cos\theta_k = -\frac{\nabla f_k^T p_k}{\|\nabla f_k\| \cdot \|p_k\|}.$$

Descent directions have $\cos\theta_k > 0$.

Theorem 2.1. *(Zoutendijk)*
Suppose every p_k is a descent direction and every step α_k satisfies the Wolfe conditions. Assume f is bounded from below and continuously differentiable on an open set N containing $\{x : f(x) \leq f(x_0)\}$, and assume ∇f is Lipschitz continuous on N with constant L. Then

$$\sum_{k=0}^{\infty} \cos^2\theta_k \|\nabla f_k\|^2 < \infty.$$

Proof.
The Armijo condition gives $f_{k+1} - f_k \leq c_1 \alpha_k \nabla f_k^T p_k$. The curvature condition rearranges to

$$\left(\nabla f_{k+1} - \nabla f_k\right)^T p_k \geq (c_2 - 1)\nabla f_k^T p_k.$$

Lipschitz continuity of the gradient bounds the same inner product by $L\alpha_k \|p_k\|^2$, so

$$\alpha_k \geq \frac{(c_2 - 1)\nabla f_k^T p_k}{L\|p_k\|^2} > 0.$$

Insert this lower bound into the decrease of f. Since $\nabla f_k^T p_k < 0$,

$$f_{k+1} - f_k \leq -\frac{c_1(1-c_2)}{L}\cos^2\theta_k \|\nabla f_k\|^2.$$

Summing the inequality yields

$$\frac{c_1(1-c_2)}{L}\sum_{k=0}^{\infty}\cos^2\theta_k\|\nabla f_k\|^2 \leq f(x_0) - \inf f < \infty.$$

■

Corollary 2.1.
Under the hypotheses of Zoutendijk's theorem, $\cos^2\theta_k\|\nabla f_k\|^2 \to 0$. If $\cos\theta_k \geq \delta > 0$ for every k, then $\|\nabla f_k\| \to 0$.

3 Gradient Descent

3.1 Convex Functions

Definition 3.1. *(Convex Set and Convex Function)*
A set D is convex if $\forall x, y \in D, \forall t \in [0,1], tx + (1-t)y \in D$. A function $f : D \to \mathbb{R}$ is convex if D is convex and

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y), \quad \alpha \in [0,1].$$

Remark.
On the interior of its domain, a convex function is differentiable almost everywhere and Lipschitz on compact subsets. If f is differentiable, then ∇f is monotone: $\langle \nabla f(x) - \nabla f(y), x - y\rangle \geq 0$, and Jensen's inequality holds.

Proposition 3.1.
Let $f : D \to \mathbb{R}$ be differentiable and convex. A point $x_0 \in D$ is a global minimizer if and only if $\nabla f(x_0) = 0$.

Proposition 3.2. *(Smoothness Inequalities)*
Suppose f is convex and differentiable and ∇f is Lipschitz continuous with constant M, namely $\nabla^2 f \preceq MI$ when the Hessian exists. Then

$$\frac{1}{2M}\|\nabla f(x) - \nabla f(y)\|^2 \leq f(y) - f(x) - \nabla f(x)^T(y-x) \leq \frac{M}{2}\|x-y\|^2,$$

and ∇f is monotone and cocoercive:

$$\frac{1}{M}\|\nabla f(x) - \nabla f(y)\|^2 \leq \langle \nabla f(x) - \nabla f(y), x - y\rangle \leq M\|x-y\|^2.$$

Remark.
If a minimizer x^* exists, the first inequality at $x = x^*$ reduces to

$$\frac{1}{2M}\|\nabla f(y)\|^2 \le f(y) - f(x^*) \le \frac{M}{2}\|y - x^*\|^2.$$

If $\nabla^2 f \preceq MI$, Taylor expansion along the gradient step gives

$$f(x_k - \alpha \nabla f_k) \le f_k - \alpha\|\nabla f_k\|^2 + \frac{1}{2}\alpha^2 M\|\nabla f_k\|^2.$$

The right-hand side is minimized at $\alpha = 1/M$, and that choice leaves

$$f_{k+1} \le f_k - \frac{1}{2M}\|\nabla f(x_k)\|^2.$$

The objective therefore decreases at every nonstationary point.

3.2 Sublinear Rate

Theorem 3.1.
Apply gradient descent with constant step $\alpha_k = \frac{1}{M}$ to a continuously differentiable convex function satisfying $\nabla^2 f \preceq MI$. If f is bounded from below, then $\nabla f(x_k) \to 0$. If at least one minimizer exists, then x_k converges to some minimizer.

Theorem 3.2. *(Sublinear Rate)*
Apply gradient descent with constant step $\alpha_k = \frac{1}{M}$ to a continuously differentiable convex function satisfying $\nabla^2 f \preceq MI$. If a minimizer x^* exists, then for every integer $k \ge 1$,

$$f(x_k) - f(x^*) \le \frac{2M\|x_0 - x^*\|^2}{k}.$$

Proof.
Write $\Delta_k = f(x_k) - f(x^*)$. The constant-step decrease and the tangent inequality $\nabla f_k^T(x_k - x^*) \ge \Delta_k$ give

$$\|\nabla f_k\| \ge \frac{\Delta_k}{\|x_k - x^*\|},$$

and therefore

$$\Delta_{k+1} \le \Delta_k - \frac{\Delta_k^2}{2M\|x_k - x^*\|^2}.$$

The distance to x^* is at most $R = \|x_0 - x^*\|$, so

$$\Delta_j - \Delta_{j+1} \geq \frac{\Delta_j^2}{2MR^2}$$

for every j. If $\Delta_k = 0$, the rate is immediate. Otherwise $\Delta_j > 0$ for all $j \leq k$, and

$$\frac{1}{\Delta_{j+1}} - \frac{1}{\Delta_j} \geq \frac{1}{2MR^2}.$$

Summing from $j = 0$ to $k - 1$ produces

$$\frac{1}{\Delta_k} \geq \frac{1}{\Delta_0} + \frac{k}{2M\|x_0 - x^*\|^2},$$

and dropping the positive initial term yields the claimed rate. ■

3.3 Strong Convexity

Definition 3.2. *(Strong Convexity)*
A differentiable function f is strongly convex with modulus $m > 0$ if $x \mapsto f(x) - \frac{m}{2}\|x\|^2$ is convex. Equivalently, when the Hessian exists, $\nabla^2 f \succeq mI$.

Proposition 3.3.
Suppose f is differentiable, strongly convex with modulus m, and ∇f is Lipschitz continuous with constant M. At the minimizer x^*,

$$\frac{1}{2M}\|\nabla f(x)\|^2 \leq f(x) - f(x^*) \leq \frac{1}{2m}\|\nabla f(x)\|^2$$

and

$$\frac{1}{M}\|\nabla f(x)\| \leq \|x - x^*\| \leq \frac{1}{m}\|\nabla f(x)\|.$$

Theorem 3.3. *(Improved Cocoercivity)*
Suppose f is differentiable, strongly convex with modulus m, and ∇f is Lipschitz continuous with constant M. Then

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{Mm}{M+m}\|x - y\|^2 + \frac{1}{M+m}\|\nabla f(x) - \nabla f(y)\|^2.$$

3.4 Linear Rate

Theorem 3.4. *(Linear Rate)*
Suppose f is differentiable, strongly convex with modulus m, and ∇f is Lipschitz continuous with constant M. If the constant gradient step satisfies $0 < \alpha \leq \frac{2}{M+m}$, then

$$\|x_k - x^*\|^2 \leq c^k \|x_0 - x^*\|^2, \quad c = 1 - \frac{2Mm\alpha}{M+m} \in [0, 1).$$

Consequently

$$f(x_k) - f(x^*) \leq \frac{M}{2} c^k \|x_0 - x^*\|^2.$$

Proof.
The iteration $x_{k+1} = x_k - \alpha \nabla f(x_k)$ expands as

$$\|x_{k+1} - x^*\|^2 = \|x_k - x^*\|^2 - 2\alpha \nabla f(x_k)^T (x_k - x^*) + \alpha^2 \|\nabla f(x_k)\|^2.$$

Improved cocoercivity at the pair (x_k, x^*), where $\nabla f(x^*) = 0$, gives

$$\nabla f(x_k)^T (x_k - x^*) \geq \frac{Mm}{M+m} \|x_k - x^*\|^2 + \frac{1}{M+m} \|\nabla f(x_k)\|^2.$$

Therefore

$$\|x_{k+1} - x^*\|^2 \leq \left(1 - \frac{2Mm\alpha}{M+m}\right) \|x_k - x^*\|^2 - \left(\frac{2\alpha}{M+m} - \alpha^2\right) \|\nabla f(x_k)\|^2.$$

The second coefficient is nonnegative precisely when $0 \leq \alpha \leq \frac{2}{M+m}$, so it may be dropped. The resulting contraction iterates to the bound on $\|x_k - x^*\|^2$. The estimate $f(x) - f(x^*) \leq \frac{M}{2} \|x - x^*\|^2$ from smoothness upgrades it to a bound on the function values. ■

Remark.
The factor c is decreasing in α on the admissible interval, so the smallest contraction available from this argument occurs at

$$\alpha = \frac{2}{M+m}, \quad c = \left(\frac{M-m}{M+m}\right)^2 = \left(\frac{\kappa - 1}{\kappa + 1}\right)^2, \quad \kappa = \frac{M}{m}.$$

4 Newton's Method

Gradient descent can be slow when the condition number κ is large. Newton's method uses the curvature of f and can converge much faster near a nondegenerate minimizer.

4.1 Pure Newton Method

Definition 4.1. *(Newton Direction)*
If $\nabla^2 f(x)$ is invertible, the Newton direction at x is

$$p = -\left(\nabla^2 f(x)\right)^{-1} \nabla f(x),$$

equivalently the solution of $\nabla^2 f(x) p = -\nabla f(x)$.

Remark.

- The local quadratic model

$$\hat{f}(x+v) = f(x) + \nabla f(x)^T v + \frac{1}{2} v^T \nabla^2 f(x) v$$

is minimized at the Newton step, so $x + p$ is the minimizer of the model.

- For a convex objective, solving $\nabla f(x) = 0$ implies $\nabla f(x) + \nabla^2 f(x) v = 0$, whose solution is again the Newton direction.
- The same p is the steepest-descent direction for the local Hessian norm $\|v\|_{\nabla^2 f(x)} = (v^T \nabla^2 f(x) v)^{\frac{1}{2}}$.

Definition 4.2. *(Pure Newton Iteration)*
Write $g_k = \nabla f(x_k)$ and $A_k = \nabla^2 f(x_k)$. If A_k is invertible, the pure Newton iteration takes the unit step $\alpha_k = 1$ along the Newton direction,

$$x_{k+1} = x_k - A_k^{-1} g_k.$$

Remark.
The iteration is reliable near a strict minimizer, where f is close to quadratic, and can overshoot or fail when x_0 is far from x^* or when A_k is not positive definite.

Theorem 4.1. *(Local Quadratic Convergence)*
Suppose f is twice differentiable and $\nabla^2 f$ is Lipschitz continuous in a neighborhood N of a point x^* in the sense that

$$\|\nabla^2 f(x) - \nabla^2 f(y)\| \le L\|x - y\|, \quad \forall x, y \in N.$$

If $\nabla^2 f(x^*) \succ 0$ and x_0 is sufficiently close to x^*, then the pure Newton iterates remain in N, converge to x^*, and both $x_k \to x^*$ and $\nabla f(x_k) \to 0$ at a quadratic rate.

Proof.
The Newton update rearranges as

$$x_{k+1} - x^* = x_k - x^* - A_k^{-1} g_k = A_k^{-1}(A_k(x_k - x^*) - g_k).$$

Taylor expansion of the gradient and $\nabla f(x^*) = 0$ give

$$A_k(x_k - x^*) - g_k = \int_0^1 (A_k - \nabla^2 f(x_k + t(x^* - x_k)))(x_k - x^*)\,\mathrm{d}t.$$

Lipschitz continuity of the Hessian bounds the integrand by $Lt\|x_k - x^*\|^2$, so

$$\|A_k(x_k - x^*) - g_k\| \le \frac{L}{2}\|x_k - x^*\|^2.$$

Since $\nabla^2 f(x^*) \succ 0$, continuity keeps A_k invertible with uniformly bounded inverse for x_k near x^*. Hence $e_{k+1} \le C e_k^2$ with $e_k = \|x_k - x^*\|$. Choosing x_0 close enough that the neighborhood is invariant and $C e_0 < 1$ yields quadratic convergence of x_k to x^*. For the gradient, $g_k + A_k p_k = 0$ with $p_k = x_{k+1} - x_k$ implies

$$g_{k+1} = \int_0^1 (\nabla^2 f(x_k + t p_k) - A_k) p_k \,\mathrm{d}t,$$

so $\|g_{k+1}\| \le \frac{L}{2}\,\|p_k\|^2 \le \frac{L}{2}\,\|A_k^{-1}\|^2 \cdot \|g_k\|^2$. Once $x_k \to x^*$, the inverse Hessians stay bounded and the same quadratic relation holds for $\|g_k\|$. ■

Example.
A strict minimizer alone does not give quadratic convergence. For $f(x) = x^4$ one has $x^* = 0$ and $f''(0) = 0$. If $x_k \neq 0$, then

$$x_{k+1} = x_k - \frac{4x_k^3}{12x_k^2} = \frac{2}{3}x_k,$$

so the iteration converges only linearly. The same loss of quadratic rate occurs at a root of higher multiplicity.

4.2 Damped Newton Method

Even under strong convexity, the unit step is typically acceptable for the Wolfe conditions only when x_k is already close to x^*. The damped Newton method keeps the Newton direction and chooses a step $\alpha_k \in (0, 1]$ by line search, most often Armijo backtracking, trying $\alpha = 1$ first.

Definition 4.3. *(Newton Decrement)*
If $\nabla^2 f(x) \succ 0$, the Newton decrement at x is

$$\lambda(x)^2 = \nabla f(x)^T \left(\nabla^2 f(x)\right)^{-1} \nabla f(x).$$

With p the Newton direction, $g = \nabla f(x)$, and $A = \nabla^2 f(x)$, there is

$$\lambda(x)^2 = p^T A p = -g^T p, \quad \hat{f}(x) - \hat{f}(x + p) = \frac{\lambda(x)^2}{2}.$$

Thus $\frac{\lambda(x)^2}{2}$ is the decrease predicted by the quadratic model.

Definition 4.4. *(Damped Newton Iteration)*
Write $g_k = \nabla f(x_k)$ and $A_k = \nabla^2 f(x_k)$. Given $x_0 \in \operatorname{dom} f$ and a tolerance $\varepsilon > 0$, compute $p_k = -A_k^{-1} g_k$ and $\lambda_k^2 = g_k^T A_k^{-1} g_k$. Stop if $\frac{\lambda_k^2}{2} \le \varepsilon$; otherwise choose α_k by Armijo backtracking and set

$$x_{k+1} = x_k + \alpha_k p_k.$$

Theorem 4.2. *(Two-Phase Convergence)*
Suppose f is strongly convex with modulus m, $mI \preceq \nabla^2 f \preceq MI$, and $\nabla^2 f$ is Lipschitz continuous with constant $L > 0$. For damped Newton with Armijo backtracking there exist $\eta \in \left(0, \frac{m^2}{L}\right)$ and $\gamma > 0$ such that

- if $\|g_k\| \ge \eta$, then $f(x_{k+1}) - f(x_k) \le -\gamma$;
- if $\|g_k\| < \eta$, then $\alpha_k = 1$ and

$$\|g_{k+1}\| \le \frac{L}{2m^2} \|g_k\|^2.$$

Proof.
Let $c_1 \in \left(0, \frac{1}{2}\right)$ be the Armijo constant and let $\beta \in (0, 1)$ be the backtracking factor. Write $d = -g^T p = p^T A p$. The Hessian bounds give $d \ge m\|p\|^2$, $d \ge \frac{\|g\|^2}{M}$, and $\|p\| \le \frac{\|g\|}{m}$. Smoothness yields

$$f(x_k + \alpha p) \le f(x_k) - \alpha d + \frac{M}{2} \alpha^2 \|p\|^2,$$

so Armijo holds for every $\alpha \le 2(1-c_1)\frac{m}{M}$. Backtracking therefore returns $\alpha_k \ge \underline{\alpha}$ with $\underline{\alpha} = \min\{1, 2\beta(1-c_1)\frac{m}{M}\}$, and

$$f(x_k) - f(x_{k+1}) \ge c_1 \alpha_k d \ge \frac{c_1 \underline{\alpha}}{M}\|g_k\|^2.$$

The first claim follows with $\gamma = \frac{c_1 \underline{\alpha}\eta^2}{M}$. Lipschitz continuity of the Hessian also gives $f(x_k+p) \le f(x_k) - \frac{d}{2} + \frac{L}{6}\|p\|^3$. The cubic remainder is at most $\left(\frac{1}{2}-c_1\right)d$ as soon as $\|p\| \le 3(1-2c_1)\frac{m}{L}$. Choose $\eta \in \left(0, \frac{m^2}{L}\right)$ small enough that this bound holds whenever $\|g_k\| < \eta$; then backtracking accepts $\alpha_k = 1$. The integral representation

$$g_{k+1} = \int_0^1 (\nabla^2 f(x_k + t p_k) - A_k) p_k \, \mathrm{d}t$$

yields $\|g_{k+1}\| \le \frac{L}{2}\|p_k\|^2 \le \frac{L}{2m^2}\|g_k\|^2$. The restriction $\eta < \frac{m^2}{L}$ keeps this region invariant under subsequent full steps. ■

Remark.
The iteration therefore has two phases. While $\|g_k\| \ge \eta$, each accepted step decreases f by at least γ, so the damped phase lasts at most $\frac{f(x_0) - \inf f}{\gamma}$ iterations. Once $\|g_k\| < \eta$, every later step is the unit Newton step, both Wolfe conditions hold, and $\|g_k\|$ converges to zero quadratically.

4.3 Hessian Modification

When A_k is not positive definite, p_k may fail to be a descent direction. A symmetric correction E_k is chosen so that

$$B_k = A_k + E_k \succ 0, \quad B_k p_k = -g_k, \quad x_{k+1} = x_k + \alpha_k p_k.$$

Then $g_k^T p_k = -p_k^T B_k p_k < 0$ whenever $g_k \neq 0$, so p_k is a descent direction and the unique minimizer of the modified quadratic model $q_k(p) = f(x_k) + g_k^T p + \frac{1}{2} p^T B_k p$. A useful correction is small, retains as much curvature as possible, and vanishes once A_k is sufficiently positive definite.

Definition 4.5. *(Modified Newton Iteration)*
Given x_0 and a tolerance $\varepsilon > 0$, the modified Newton iteration repeats: stop if $\|g_k\| \le \varepsilon$; form and factor $B_k = A_k + E_k \succ 0$, taking $E_k = 0$ when A_k already satisfies the chosen safeguards; solve $B_k p_k = -g_k$; try $\alpha_k = 1$ first and otherwise take a Wolfe step; set

$$x_{k+1} = x_k + \alpha_k p_k.$$

Definition 4.6. *(Bounded Modified Factorization)*
The condition number of a modified Hessian is $\kappa_2(B_k) = \frac{\lambda_{\max}(B_k)}{\lambda_{\min}(B_k)}$. A modification rule has the bounded modified factorization property if bounded input Hessians produce matrices B_k with $\kappa_2(B_k)$ uniformly bounded.

Theorem 4.3.
Let $f \in C^2(\mathbb{R}^n)$ and suppose the level set $\{x : f(x) \le f(x_0)\}$ is compact. Assume the modified Newton iteration uses Wolfe steps with fixed constants $0 < c_1 < c_2 < 1$ and that $B_k \succ 0$ with $\kappa_2(B_k) \le K$ independently of k. Then the algorithm either stops at a stationary point or satisfies $\|g_k\| \to 0$. In particular every accumulation point of an infinite sequence is stationary.

Proof.
Compactness of the level set and $f \in C^2$ supply the Lipschitz hypothesis of Zoutendijk's theorem. The angle bound $\cos\theta_k \ge \frac{1}{K}$ and the corresponding corollary yield $\|g_k\| \to 0$. ■

Remark.
If $x_k \to x^*$ with $\nabla f(x^*) = 0$, $\nabla^2 f(x^*) \succ 0$, and $E_k = 0$ for all large k, the direction becomes the Newton direction. With $c_1 < \frac{1}{2}$ and a line search that tries the unit step first, both Wolfe inequalities hold for small p_k, so $\alpha_k = 1$ is eventually accepted and local quadratic convergence applies.

Drop the iteration index and write $A = \nabla^2 f(x)$. Four standard constructions of $B = A + E$ follow.

- *Eigenvalue modification.* Write $A = Q\Lambda Q^T$ and raise every eigenvalue below a floor $\delta > 0$,

  $$B = Q\,\mathrm{diag}(\max\{\lambda_i, \delta\})Q^T.$$

  Then $B \succeq \delta I$, and B is the unique nearest matrix to A in the Frobenius norm among symmetric matrices bounded below by δI. If $\|A\| \le M_A$, then $\kappa_2(B) \le \frac{\max\{M_A, \delta\}}{\delta}$. A full eigendecomposition is costly and can destroy sparsity.
- *Multiple of the identity.* Take $B = A + \tau I$ with $\tau = \max\{0, \delta - \lambda_{\min}(A)\}$. For large τ the direction approaches a scaled gradient step. In practice one need not compute $\lambda_{\min}(A)$: a Cholesky retry starts from $\tau = 0$ and doubles τ until $A + \tau I$ factors.
- *Modified Cholesky.* In an LDL^T factorization, after forming the residual column set $\theta_j = \max_{i>j}|c_{ij}|$ and

$$d_j = \max\left\{|c_{jj}|, \left(\frac{\theta_j}{\beta}\right)^2, \delta\right\}.$$

This keeps $d_j \geq \delta$ and $|\ell_{ij}\sqrt{d_j}| \leq \beta$, and produces $B = LDL^T = A + E$ with E diagonal and nonnegative. The correction must be applied during the factorization; for fixed dimension and positive parameters, bounded A yields bounded $\kappa_2(B)$.

- *Symmetric indefinite factorization.* Symmetric pivoting produces $PAP^T = LDL^T$, where D is block diagonal with blocks of size 1 or 2. Diagonalize each block and raise its eigenvalues to at least δ, giving $\hat{D} \succeq \delta I$ and

$$B = P^T L \hat{D} L^T P = A + E \succ 0.$$

Only tiny eigenproblems are solved, but E need not be diagonal, and $\hat{D} \succeq \delta I$ does not imply $B \succeq \delta I$: one has $\kappa_2(B) \leq \kappa_2(L)^2 \kappa_2\big(\hat{D}\big)$.

4.4 Quasi-Newton Methods

Definition 4.7. *(Quasi-Newton Iteration)*
Quasi-Newton methods replace A_k by a symmetric positive definite approximation B_k and solve

$$B_k p_k = -g_k, \quad x_{k+1} = x_k + \alpha_k p_k.$$

Then p_k is a descent direction and minimizes $m_k(p) = f(x_k) + g_k^T p + \frac{1}{2} p^T B_k p$.

Theorem 4.4. *(Accurate Directions and Unit Steps)*
Let f be twice continuously differentiable near x^*. Fix Wolfe parameters $0 < c_1 < \frac{1}{2}$ and $c_1 < c_2 < 1$. Suppose a descent iteration $x_{k+1} = x_k + \alpha_k p_k$ satisfies the Wolfe conditions and converges to x^*, where $\nabla f(x^*) = 0$ and $A_* := \nabla^2 f(x^*) \succ 0$. If $\frac{\|r_k\|}{\|p_k\|} \to 0$, then the unit step $\alpha_k = 1$ is admissible for all large k. If the line search tries $\alpha = 1$ first and accepts it whenever it is admissible, then eventually $\alpha_k = 1$ and the convergence is superlinear, namely

$$\frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} \to 0.$$

Proof.
Continuity and $A_* \succ 0$ give $mI \preceq A_k \preceq MI$ for large k. Write $\|r_k\| \leq \varepsilon_k \, \|p_k\|$ with $\varepsilon_k \to 0$. From $A_k p_k = -g_k + r_k$ one obtains $(m - \varepsilon_k) \, \|p_k\| \leq \|g_k\|$. Local boundedness of the Hessian and $\nabla f(x^*) = 0$ give $\|g_k\| = O(\|x_k - x^*\|)$, so $\|p_k\| = O(\|x_k - x^*\|)$ and $p_k \to 0$. Taylor expansion is therefore valid along the segments

from x_k to $x_k + p_k$. Set $q_k = p_k^T A_k p_k \geq m \, \|p_k\|^2$. The residual identity yields $g_k^T p_k = -q_k + o(\|p_k\|^2)$, and

$$f(x_k + p_k) - f(x_k) - c_1 g_k^T p_k = -\left(\frac{1}{2} - c_1\right) q_k + o(\|p_k\|^2) < 0$$

for large k. This is the Armijo condition; the hypothesis $c_1 < \frac{1}{2}$ supplies the margin that absorbs the remainder. Another expansion gives $\nabla f(x_k + p_k) = g_k + A_k p_k + o(\|p_k\|) = o(\|p_k\|)$, while $-g_k^T p_k \geq \frac{m}{2} \|p_k\|^2$, so

$$\frac{|\nabla f(x_k + p_k)^T p_k|}{-g_k^T p_k} \to 0.$$

The strong Wolfe curvature inequality follows, and with it the ordinary curvature inequality. The stated line search therefore eventually takes the unit step. Local boundedness of A_k^{-1} gives $p_k - p_k^N = A_k^{-1} r_k = o(\|p_k\|) = o(\|x_k - x^*\|)$. Writing $h_k = x_k - x^*$ and using the fundamental theorem of calculus,

$$x_k + p_k^N - x^* = A_k^{-1} \int_0^1 (A_k - \nabla^2 f(x^* + t h_k)) h_k \, \mathrm{d}t = o(\|h_k\|).$$

Once $\alpha_k = 1$,

$$x_{k+1} - x^* = \left(x_k + p_k^N - x^*\right) + \left(p_k - p_k^N\right) = o(\|x_k - x^*\|).$$

■

Definition 4.8. *(Dennis-Moré Condition)*
A sequence of matrices $\{B_k\}$ satisfies the Dennis-Moré condition along $\{p_k\}$ if

$$\frac{\|(B_k - A_*) p_k\|}{\|p_k\|} \to 0.$$

For a quasi-Newton direction one has $r_k = (A_k - B_k) p_k$, so once $A_k \to A_*$ the residual condition is equivalent to this estimate.

Theorem 4.5. *(Dennis-Moré)*
Suppose f is twice continuously differentiable near a stationary point x^* with $A_* \succ 0$. Let $x_{k+1} = x_k + p_k$ with $B_k p_k = -g_k$, where B_k is symmetric positive definite, and assume $x_k \to x^*$. The convergence is superlinear if and only if the Dennis-Moré condition holds.

Proof.
Sufficiency follows from the residual condition and the comparison with the Newton step in the previous proof, applied to this full-step sequence. Conversely, suppose the convergence is superlinear and write $h_k = x_k - x^*$. Then $p_k = h_{k+1} - h_k$, so $h_{k+1} = o(\|h_k\|)$, $p_k = -h_k + o(\|h_k\|)$, and $\frac{\|p_k\|}{\|h_k\|} \to 1$. Differentiability of the gradient gives $g_k = A_* h_k + o(\|h_k\|)$, and therefore

$$(B_k - A_*)p_k = -g_k - A_* p_k = -A_* h_{k+1} + o(\|h_k\|) = o(\|p_k\|).$$

■

Example.
Directional accuracy need not imply matrix convergence. For

$$f(x) = \frac{1}{2}(x_1^2 + x_2^2) + \frac{1}{4}x_1^4, \quad x^* = (0,0), \quad A_* = I,$$

take $B_k = \operatorname{diag}(1, 2)$ at every iteration. Starting from $x_0 = (a, 0)$ with $0 < |a| < 1$, full steps produce $x_k = (a_k, 0)$ with $p_k = \left(-a_k - a_k^3, 0\right)$ and $a_{k+1} = -a_k^3$. Thus $a_k \to 0$ cubically and $(B_k - A_*)p_k = 0$, even though $\|B_k - A_*\| = 1$ for every k. The matrix is exact in the direction used by the iteration.